



Detection of abandoned objects based on YOLOv9 and background differencing

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Abstract

Abandoned objects on highways can cause serious traffic accidents. Abandoned objects detection algorithms are of great significance in reducing traffic accidents and improving the efficiency of highway management. Existing traditional image processing detection algorithms are difficult to detect small targets in the complex and changing highway environment. And the current deep learning methods are limited by the number of samples and the influence of vehicle occlusion resulting in lower detection accuracy. Aiming at the above problems, a abandoned object detection algorithm for highway is designed by combining the background modeling algorithm MOG2 and the YOLOv9 detector. Utilizing the difference of the two algorithms on the recognition mechanism of abandoned objects, the results of the YOLOv9 detector are used as a guide to make a comprehensive judgment on the low confidence targets combined with the iterative difference results of MOG2. The actual test results on highways show that the abandoned objects detection algorithm based on YOLOv9 and MOG2 has an improved precision and recall rate for abandoned objects detection compared to a single detection algorithm, with an average precision of 86.7% and a recall rate of 89.6%. The algorithm effectively complements the existing abandoned objects detection algorithms and has engineering practical value.

Keywords Abandoned object detection · Computer vision · YOLO algorithm · Gaussian mixture model

1 Introduction

Traffic safety on highways has always been a hot issue of concern, and abandoned objects on the road is one of the main reasons for the threat to road safety. Most of the abandoned objects on the highway are luggage, goods, spare tires, or road debris from traffic accidents that are not properly placed on the vehicle. The collision of high-speed vehicles with abandoned objects can lead to serious traffic accidents. However, most of the cities still rely on manual road monitoring and field patrol by road maintenance vehicles to detect

abandoned objects. This process is labor-intensive and time-consuming, and the detection accuracy is low. Therefore, it is of great significance and application value to study the automatic detection algorithm of highway abandoned objects. The current automatic highway abandoned objects detection algorithms can be divided into two main categories: background model-based abandoned objects detection algorithms [1, 2] and deep learning based abandoned objects detection methods [3].

The background model based abandoned object detection algorithm analyzes the current frame in an image sequence by constructing a clean and constantly updated background image and comparing or varying it with the background image. It calculates the regions where the difference exceeds a certain threshold and recognizes them as target objects.

Currently, deep learning-based target detection algorithms can be categorized into two groups, one is regression-based single-stage (one-stage) algorithms, such as Centernet [4], RetinaNet [5], SSD series [6], YOLO series [7–9], etc., and the other is two-stage algorithms based on region suggestion, such as R-CNN [10], Fast R-CNN [11], Faster R-CNN [12] and so on. At present, the YOLO series models

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with better performance have been widely used in detection tasks in engineering fields [13, 14]. The YOLO algorithm achieves promising results in the detection of abandoned objects on highways under experimental conditions. However, in the real world, there is a wide variety of abandoned objects on highways and the training dataset cannot cover all types of debris. As a result, deep learning-based methods exhibit poor algorithmic generalization for untrained small targets. Aiming at the above problems, this paper combines the background modeling algorithm MOG2 [15, 16] and YOLOv9 [17] detector, which guarantees the speed and accuracy of abandoned Object Detection Algorithm for Highway while reducing the computational cost, and effectively avoids misjudgment due to the target occlusion or road surface interference. At the same time, data from different highway scenes are selected for training, which improves the generalization ability of the algorithm.

2 Theoretical background

2.1 Dynamic background modeling

Dynamic background modeling is an algorithm that dynamically builds a scene background model according to the scene changes. Dynamic background modeling algorithms based on Gaussian Mixture Model (GMM) have been shown to be more effective in complex environments [18]. The core of the algorithm is a background modeling method based on Gaussian function probability density information. In real-world scenarios, multiple Gaussian models are used to describe pixel values that may be background to create a more accurate background model. Mixed Gaussian background modeling focuses on modeling the pixel values in an image through a mixed Gaussian distribution model, which uses a mixed Gaussian distribution model with M distributions as shown in Eq. 1. BG and FG are identifiers that identify pixels as belonging to the background and foreground, respectively. In order to adapt to the disturbances in the picture, a suitable time window T is chosen, so the sample set $\chi_T = \{x^{(t)}, \dots, x^{(t-T)}\}$ is obtained at time t . For each new sample χ_T is updated and $p(x | \hat{\chi}, BG)$ is re-estimated, the sample set contains the pixel values of the foreground region.

$$\hat{p}(x | \chi_T, BG + FG) = \sum_{m=1}^M \hat{\pi}_m N\left(\hat{x}; \hat{\mu}_m, \hat{\sigma}_m^2 I\right) \quad (1)$$

where $\hat{\mu}_1, \dots, \hat{\mu}_M$ is the Gaussian distribution mean estimate and $\hat{\sigma}_1^2, \dots, \hat{\sigma}_M^2$ is the Gaussian distribution variance estimate. The covariance matrix is assumed to be a focusing matrix, the unit matrix I has appropriate dimensions, and the

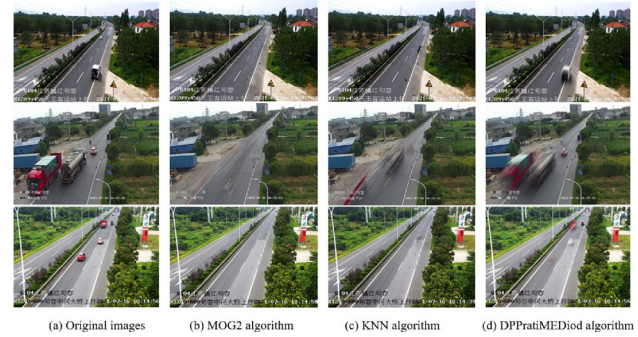


Fig. 1 Figure of background modeling results of three algorithms in different scenarios

$\hat{\pi}_m$ representation has non-negative weights and sums to one. After establishing the initial background model, the parameters in the background model are continuously updated by selecting certain weighting coefficients according to a certain number of reference frames.

To identify an appropriate background modeling algorithm for the recognition of abandoned objects, we conducted comparative experiments using the following methods: the MOG2 algorithm, which is considered superior within the Gaussian Mixture Model (GMM) framework; the KNN-based Dynamic Background Modeling algorithm [16]; and the DPPratiMEDiod algorithm from the Background Subtraction Library (BGS) [19]. The results of the MOG2 algorithm with the other two Dynamic background modeling algorithms are given in turn in Fig. 1. In videos with simple background, the three algorithms build high quality background images. However, only the MOG 2 algorithm achieved better results in videos with complex backgrounds and more vehicles. In summary, this paper uses the MOG2 algorithm for Dynamic background modeling of video, selecting the video after 300 frames to record the modeling results once. A series of background images without vehicles can be generated after Dynamic background modeling.

2.2 YOLOv9 object detection algorithm

The YOLOv9 algorithm, renowned for its high performance in object detection, is utilized in this section for recognizing abandoned objects. It mitigates information loss common in deep learning with the introduction of Programmable Gradient Information (PGI) and a novel lightweight architecture, Generalized Efficient Layer Aggregation Network (GELAN). This combination enhances YOLOv9's detection capabilities, surpassing current real-time detectors on the MS COCO dataset.

YOLOv9's architecture, as depicted in Fig. 2, includes a backbone for feature extraction, a head for prediction, and auxiliary components for training optimization. The backbone, comprising Conv and RepNCSPELAN4 modules,

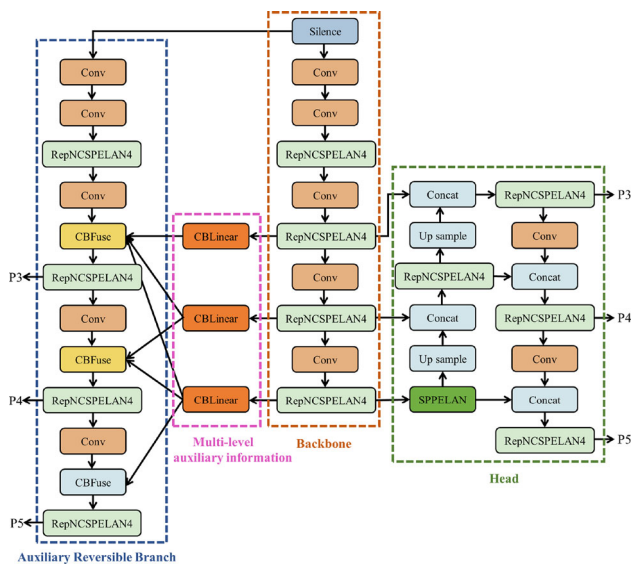


Fig. 2 YOLOv9 network structure diagram

enhances feature extraction through local aggregation and global integration. The head, with modules like UPSample and SPPELAN, fuses features and predicts bounding boxes and categories. Auxiliary components improve backbone performance during training. YOLOv9 comes in five versions, varying in network size: YOLOv9-T, YOLOv9-S, YOLOv9-M, YOLOv9-C, and YOLOv9-E.

3 Methods

Aiming at the problems of misdetection and omission caused by small target size, target occlusion, and environmental noise interference in the abandoned objects detection task, a highway abandoned objects detection algorithm based on deep learning with Iterative Difference of Background Modeling (IDBM) is designed. First, Dynamic background modeling is performed using MOG2 to obtain background information. In the method of IDBM, the detection of the abandoned objects entering the stationary state is carried out. Then, a series of background images obtained by the above method are fed into the YOLOv9 network for the detection of abandoned objects, and a reasonable confidence threshold is set to judge the abandoned objects detected by deep learning; finally, for the suspected abandoned objects with low confidence, a new method of detecting abandoned objects with higher accuracy is proposed by combining the detection results of YOLOv9 and the background modeling target detection results.

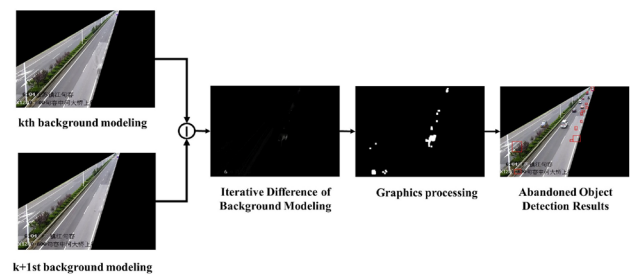


Fig. 3 IDBM algorithm flowchart

3.1 Iterative difference of background modeling algorithm

Dynamic background modeling is concerned with the background situation of a roadway over time, which includes a abandoned object target becoming part of the background at rest after hitting the ground. This study bypasses the analysis of the abandoned process of debris and instead focuses on investigating the stationary characteristics of objects on the road surface. Based on this, the paper proposes a detection method for abandoned objects using the IDBM algorithm. The method employs continuous dynamic background modeling to integrate the abandoned object, once it reaches a stationary state, into the background image. Concurrently, it conducts a differential operation on the adjacent background images preceding and following the grounding of the abandoned object to discern the abandoned object target.

According to the method shown in Fig. 3, the algorithm first obtains a mask of the suspected abandoned objects by differencing the background images created at the (k+1st) and (kth) times. Next, binarization of these differences and corrosion expansion of the images are performed to further accentuate the features of the abandoned objects. Finally, the outer rectangle of each irregular suspected abandoned object target is defined as the location of the abandoned object. After differential and graphical manipulation, the abandoned object targets are clearly visible.

3.2 Improved YOLOv9 for abandoned objects detection

The regression loss function in the target detection algorithm plays an important role in evaluating and optimizing the performance. The regression loss function used in YOLOv9 is CIOU (Complete Intersection over Union) loss and its formula is shown below:

$$CIOU = IoU - \frac{\rho^2(b, b^{gt})}{c^2} - \alpha v$$

$$L_{CIOU} = 1 - CIOU$$
(2)

b and b^{gt} denote the centroids of the prediction and truth frames, respectively, $\rho^2(\bullet)$ is used to compute the Euclidean distance between the two points, and c denotes the diagonal length of the minimum bounding box of the prediction and truth frames. However, CIoU has some defects, v The aspect ratio describes the relative value, which has some ambiguity. The problem of difficult and easy sample balance is not considered, which leads to the weak generalization performance of the model.

Most of the current improved loss functions accelerate the convergence by adding new loss terms, and do not pay attention to the limitations of the IoU itself. Inner-IoU[20] proposes to compute the IoU with auxiliary bounding box to improve the generalization ability, and the specific computation process is shown in Eqs. (5–9), and the scale factor ratio is used to control the size of the auxiliary bounding box.

$$b_l = x_c - \frac{w \times ratio}{2}, b_r = x_c + \frac{w \times ratio}{2} \quad (3)$$

$$b_t = y_c - \frac{h \times ratio}{2}, b_b = y_c + \frac{h \times ratio}{2} \quad (4)$$

Through Eqs. (5) and (6), the center point of the detection frame is transformed to obtain the corner vertices of the auxiliary detection frame. The prediction frame and the real frame output from the model are transformed accordingly, and the calculation results of the real frame and the prediction frame are expressed by b^{gt} and b^{pred} , respectively.

$$inter = \left(\min(b_r^{gt}, b_r) - \max(b_l^{gt}, b_l) \right) \times \left(\min(b_b^{gt}, b_b) - \max(b_t^{gt}, b_t) \right) \quad (5)$$

$$union = (w^{gt} \times h^{gt}) \times (ratio)^2 + (w \times h) \times (ratio)^2 - inter \quad (6)$$

$$IoU^{inner} = \frac{inter}{union} \quad (7)$$

As shown in Eqs. (5–7), inner-IoU actually computes the IoU between the auxiliary margins. where $ratio \in [0.5, 1.5]$, when $ratio < 1$, the auxiliary margins are smaller than the actual margins, resulting in a regression whose effective range is smaller than the IoU loss, but whose gradient is larger in absolute value than that obtained from the IoU loss, which helps to accelerate the convergence of high convergence of the IoU samples. And when $ratio > 1$, the auxiliary margin is larger than the actual frame, which enlarges the effective range of the regression, which is beneficial for the regression of low IoU samples. Since there is only a scale difference between the auxiliary border and the actual border, the trend of its IoU value during the regression process is consistent with the trend of the IoU value of the actual border, which can reflect the quality of the regression results of the actual border.

MPDIoU [21] is a loss function for a novel bounding box similarity comparison metric based on minimum point distances, which incorporates factors such as overlapping or non-overlapping areas, centroid distances, and deviations in width and height to improve convergence speed.

$$MPDIoU = IoU - \frac{\rho^2(P_1^{pred}, P_1^{gt})}{w^2 + h^2} - \frac{\rho^2(P_2^{pred}, P_2^{gt})}{w^2 + h^2} \quad (8)$$

Among them, P_1^{pred} , P_2^{pred} , P_1^{gt} , P_2^{gt} refer to the points at the upper-left and lower-right corners of the prediction frame and the real frame, respectively, and $\rho^2(P_1^{pred}, P_1^{gt})$ calculates the distance between the corresponding points. The idea of inner-IoU is used to transform the MPDIoU by replacing the CIoU in it to ensure that the network converges faster during the training process and obtains higher regression localization accuracy.

3.3 Detection of abandoned objects based on YOLOv9 and IDBM

The flow of the algorithm is shown in Fig. 4, which is mainly divided into three stages:

- Step 1 Reads the surveillance camera video sequence and performs Dynamic background modeling for each frame and records the results every 300 frames. Then use Iterative Difference with Graphics Processing to obtain suspicious abandoned objects.
- Step 2 Feeds the background image generated in the first stage into the YOLOv9 target detection model for the recognition of abandoned objects. Detecting and localizing abandoned targets in an image by extracting image features.
- Step 3 Finally, the detection results of the first two stages are combined. When the confidence level of the deep learning detected abandoned objects is greater than a certain threshold is directly recognized as abandoned objects; When the confidence level falls below the predetermined threshold, the assessment is deemed inadequate. Subsequently, the coordinate and intersection ratio is calculated between the detection results of IDBM and the low confidence target of YOLOv9, and the threshold value is set to determine whether the target is a dropped object. The threshold is a specific value based on several experiments.

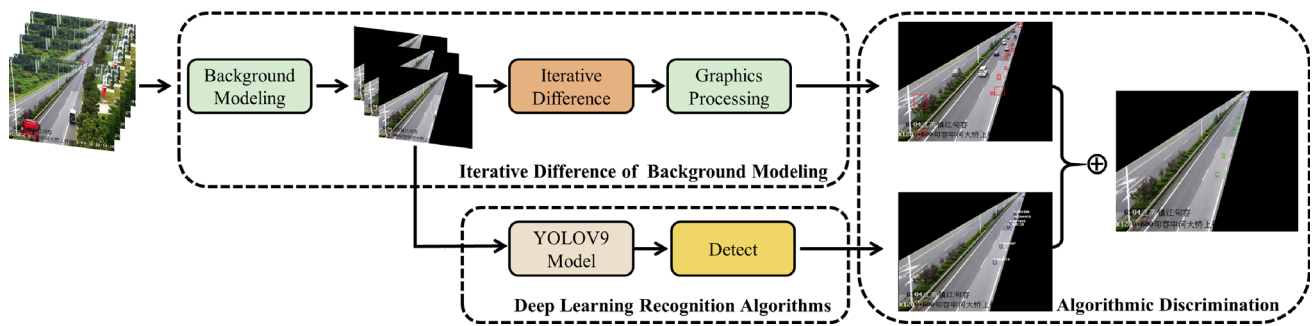


Fig. 4 Detection of abandoned objects on highways Algorithmic flow

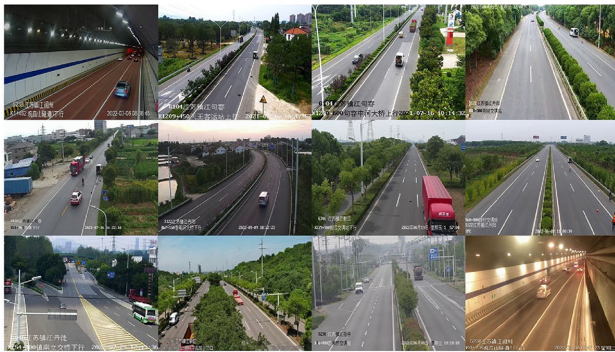


Fig. 5 Screenshot of traffic monitor

4 Results and analysis

4.1 Experimental data and experimental platform

The experimental data in this paper comes from the video surveillance material provided by a highway business development center in Jiangsu Province, China, and the surveillance cameras are all of 1280*960 resolution. Through the NVR hard disk recorder to obtain video clips containing the whole process of abandoned objects falling. As shown in Fig. 5, these are surveillance images recorded by some surveillance cameras at different time periods, different road scenes and different angles. In this paper, experiments are conducted using human-site simulated highway abandoned objects surveillance videos to evaluate the effectiveness of the algorithm.

The hardware of the experimental and testing platform for the abandoned object detection algorithm is a computer with Intel(R) Core(TM) i5-12,400 CPU, 16G, and NVIDIA GTX1060, and the model training server is a server with Intel(R) Xeon(R) Gold 5120, 64G, and 2*NVIDIA Tesla V100 32 G. In this paper, a total of 2813 images of abandoned objects from surveillance video footage and web footage were collected and formed the training set and test set in the ratio of eight to two. YOLOv9-C is chosen as the model for abandoned objects detection in this paper, and the inner-

MPDIou loss function is used to train the model for 300 rounds before it is used for recognition test.

4.2 Evaluation Metricst

In industrial environments, the primary focus of abandoned objects detection algorithms is to provide fast detection within a certain timeframe after the abandoned objects have fallen. This means that while the algorithms do not necessarily have to detect the surveillance video in real time, effectively minimizing false alarms is critical to the reliability of a long-term abandoned object detection system. To evaluate the performance of our abandoned objects detection algorithm, we focus on two key metrics: Precision (AP) and Recall (AR). The formulae are expressed as shown in 11 below:

$$\text{Precision} = \frac{TP}{(TP + FP)}$$

$$\text{Recall} = \frac{TP}{(TP + FN)}$$
(9)

where TP denotes the number of true-positive tests, FP denotes the number of false-positive tests, TN denotes the number of true-negative tests, and FN denotes the number of false-negative tests.

4.3 YOLOv9 improved effect comparison experiment

In order to evaluate the detection performance of the improved YOLOv9 model on high-speed sprinkles, ablation experiments with different optimization algorithms were conducted on the test set. The results are shown in Table 1.

The experimental results show that when MPDIoU is used instead of CIoU in original YOLOv9, the detection accuracy is improved by 2.32% when other network structures are unchanged. InnerCIoU improves the performance of YOLOv9 by 2.38%. The improved YOLOv9 detection algorithm proposed in this paper integrates the above two improvement strategies, and the accuracy is improved by 2.75% compared with the original YOLOv9.

Table 1 YOLOv9 algorithm improved comparative experiment

Method	P	R	mAP@0.5
YOLOv9	0.9108	0.9406	0.9361
YOLOv9+MPDIoU	0.9399	0.9174	0.9578
YOLOv9+InnerCIoU	0.9395	0.9119	0.9584
Ours	0.9425	0.9209	0.9619

4.4 Explanation of comparative algorithms

In order to verify the effectiveness of the algorithm in this paper, three sets of surveillance videos with different scenes simulating vehicle object throwing are used as the experimental datasets. The precision (AP) and recall (AR) of the experimental results were recorded and compared using the IDBM algorithm, YOLOv9, YOLOv8 and RT-DETR algorithms as the comparison algorithms for the algorithms in this paper, respectively.

For the IDBM algorithm, Dynamic background modeling is performed using the MOG2 algorithm. The background model is updated every 300 frames for the video sequence, and a difference operation is performed between neighboring background reference models to obtain the abandoned objects targets.

For the YOLOv9 algorithm, we choose the YOLOv9-C model for comparative analysis. YOLOv8 and RT-DETR select YOLOv8-L and RT-DETR-R50 with similar params scales as YOLOv9-C. The model was configured as follows: conf=0.25, iou=0.45, cuda device=1, and trained for a total of 300 epochs. all other parameters were set to default values.

4.5 Analysis of results

Figure 6 shows the detection results of our proposed method on multiple test datasets. In the figure: each row shows the output images of different target detection algorithms for the same test video. Each column shows the output images of different target detection algorithms for different test videos. Further note: Column (1) shows the original image sequence. Column (2) shows the detection results of the IDBM algorithm with abandoned objects labeled in red boxes. Columns (3) to (5) show the detection results of YOLOv9, YOLOv8 and RT-DETR with the abandoned objects labeled in red boxes. Column (6) shows the comparison results of our algorithms, where the abandoned objects are labeled with green boxes.

4.6 Comparative evaluation

First, the surveillance videos of the three scenes are pre-processed with highway region selection simultaneously to

reduce the environmental interference. The test results of IDBM algorithm are shown in column (2). It can be seen from the result graph that the main reason for misjudgment is that the open scene is affected by the natural environment or the slow movement of vehicles, which leads to a large amount of noise being misjudged as abandoned targets. The test result graph of the YOLOv9 algorithm is shown in column (3). From the result graph, it can be seen that the YOLOv9 algorithm produces different recognition confidence levels depending on whether the type of abandoned object has been trained or not, with a maximum of more than 0.8 and a minimum of about 0.33. noise. the detection results of the YOLOv8 algorithm and the RT-DETR algorithm are shown in the graphs of Columns (4) and Columns (5). These two algorithms detect many low confidence targets. In practical engineering, such low confidence targets are judged as noise for missed detection cases.

The test results of the algorithm in this paper are shown in column (6) of Fig. 6. The figure shows the better detection of abandoned objects on the highway by the algorithm of this paper. It is able to better identify targets such as tires, stools, triangular cones, boards, etc. on the road, and the detection frame position is accurate and there is no overlapping phenomenon. A total of 24 abandoned objects were detected in the above frame sequences. After manual review of the images captured by the surveillance video, the actual total number of abandoned objects is 25, and there is one missed detection of abandoned objects. The reason for the omission is that the color of the abandoned objects is close to the road surface, and the deep learning method can not accurately identify this situation, but the probability of this situation is low in practice. Through the experiment, it can be shown that the algorithm in this paper is accurate and effective for the detection of abandoned objects.

As described below, Table 1 presents a comparative analysis of the performance metrics of these algorithms on the test set. Insightful observations from these results show that while IDBM, which is centered on background modeling, exhibits impressive Recall(AR) on the test. However, it also suffers from low accuracy (AP), which reduces its usefulness in real-world industrial applications. In contrast, the YOLOv9, YOLOv8 and RT-DETR has achieved an AP of about 80% in the test, but nearly half of the abandoned objects did not meet the threshold criteria to determine whether they were abandoned objects or ground stains, shadows, or picture noise, resulting in missed detections.

Table 1 shows that the highway discard body detection algorithm based on YOLOv9 and background difference proposed in this paper has obvious advantages. Compared with the abandonment body detection method based on IDBM algorithm, this method can eliminate the background noise in a large amount, thus reducing the false detection rate and improving the accuracy index (Table 2). Compared with the

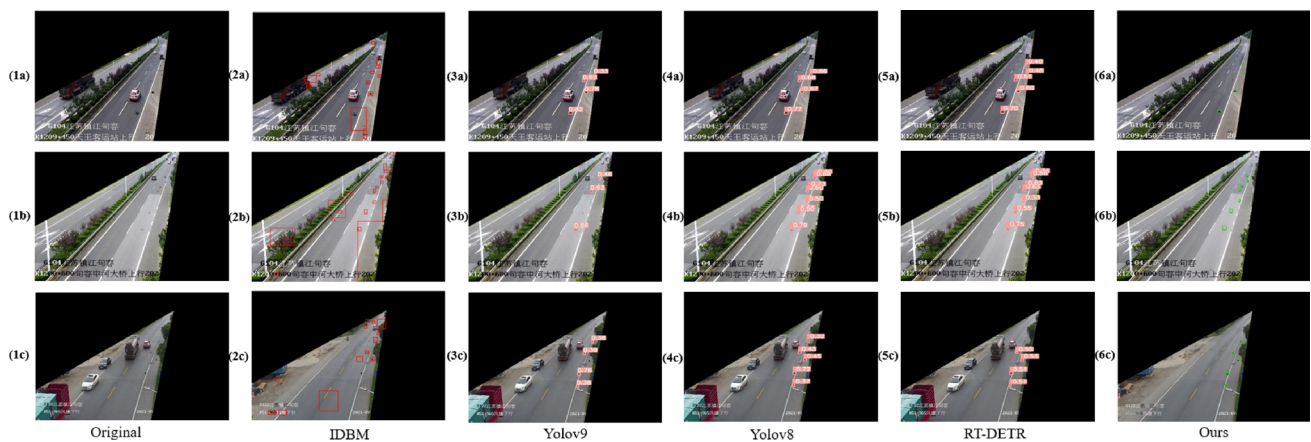


Fig. 6 Test results for three specific scenarios

Table 2 Detection results of abandoned objects by different algorithms

Project	Scene 1		Scene 2		Scene 3	
	AP	AR	AP	AR	AP	AR
IDBM	8.03%	76.80%	10.86%	73.91%	13.85%	79.35%
YOLOv9	86.67%	56.52%	79.42%	58.70%	80.28%	61.96%
YOLOv8	86.04%	53.62%	78.12%	54.35%	80.88%	59.78%
RE-DETE	88.37%	55.07%	78.79%	56.52%	79.45%	63.04%
Ours	87.14%	88.41%	87.23%	89.13%	85.71%	91.30%

abandonment detection method based on YOLOv9, YOLOv8 and RT-DETR algorithm, the algorithm in this paper can overcome the problems of limited data set, different morphology and difficult to extract features, thus reducing the leakage detection rate and improving the recall rate index. The actual test results on the highway show that the high-speed abandonment detection algorithm based on YOLOv9 and MOG2 improves the precision and recall of abandonment detection compared with the single detection algorithm, with an average precision of 86.7% and a recall of 89.6%.

5 Conclusion

Abandoned objects detection is a key part of building a smart highway, and it is also a hot issue in the research of this field in recent years. In this paper, for the hot issue of abandoned objects recognition, the mechanism of abandoned objects generation is deeply studied, and the shortcomings of traditional image processing methods and deep learning methods are analyzed. In view of these problems, this paper proposes a highway abandonment detection algorithm with YOLOv9 and Background Differencing. The experimental results show that the algorithm proposed in this paper has excellent detection results for all types of abandoned objects. The performance of the proposed model depends on the YOLOv9 detector. When the accuracy of YOLOv9 detec-

tor in processing unknown samples was reduced, we adopted IDBM algorithm for supplementary judgment to reduce the occurrence of missed detection. In the future, we will further optimize YOLOv9 and collect the abandoned objects data to improve the detection accuracy.

Author Contributions Huajun Song Conceptualization, Methodology, Investigation, Writing-Original Draft Jinbo Wang Data Curation, Writing-Original Draft, Formal Analysis Yunze Zhang Resources, Formal Analysis, Writing-Original Draft

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Data availability The data utilized in this study are not publicly available due to restrictions imposed by the third-party organization from which they were obtained.

Declarations

Conflict of interest The authors have no conflict of interest to declare that are relevant to the content of this article.

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